

Scheme for Internal Assessment (IA)

Branch	EXTC	Year	B. Tech. T.Y.
Academic year:	July 2026- October26	Course Code and Name	216U03C501 / Digital Communications
Semester:	V		
Faculty In-charge	Dr. Bharati Singh		

Distribution of 20 Marks	The student will be evaluated based on two tasks of 20 marks each. If any of the tasks given is not completed / submitted / shown / evaluated then the marks assigned for that task will be zero. IA Scheme: I. Open note book test: If remained absent on the day for the test without approved valid reason, retest will be conducted but maximum marks will be 15 marks. II. Mini project: Groups of 4 to 5 students will design, simulate and give the presentation/demonstration of a simple digital communication based system and demonstrate AI Integration in Digital Communication for one of the 5 AI use case. The groups and choice of AI use case will be decided by first week of August. Students will be ready with the project and presentation by October 19th 2026. Average marks of the two tasks will be considered.		
Sr. No.	Task	Description of task	Schedule
1	Open note book test	Questions on Module 1, 2, 3.1, 3.2	3rd week of September
2	Develop a mini project to show AI Integration in Digital Communication for one of the 5 AI use case	Groups of 4 to 5 students will design, simulate and give the presentation/demonstration of a simple digital communication based system and demonstrate AI Integration in Digital Communication for one of the 5 AI use case. The groups and choice of AI use case will be decided by first week of August.	In 3rd or 4th week of October

Final IA marks will be out of 20 Average of two IA components listed above

Test syllabus

	Topics	CO Mapping
ISE	Module No.: 1, 2, 3.1, 3.2	CO1,CO2,CO3

Module No.	Unit No.	Contents
1	Sampling Process and types of sampling	
	1.1	Sampling theorem for low pass & band pass signal, Ideal sampling process, Spectrum of sampled signal, Reconstruction of sampled signal and amplitude response of reconstruction filter.(No derivation)
	1.2	Signal distortion in sampling, signal-to-distortion Ratio, practical aspect of sampling, flat top sampling, spectrum of flat top sampled signal.

2	Information Theory , Source coding and Channel capacity	
	2.1	Basic Signal Processing Operations in Digital Communications, Uncertainty, Information, and Entropy
	2.2	Source Coding Theorem, Shannon Fano source coding, Huffman source coding , Lempel –Ziv
	2.3	Channel capacity, Channel coding theorem, Channel Capacity Theorem
3	Baseband Modulation and Demodulation Techniques	
	3.1	Signal representation: GRAM-SCHMIDT Orthogonalization procedure, Geometric representation of signals
	3.2	Baseband data formats and their power spectral density: NRZ unipolar, NRZ polar, NRZ bipolar, Manchester, M-ary PAM signals (derivations not expected)

Date: 15th July 2026

Signature of Faculty In-charge of Course

Dr. Bharati Singh